Wood Design

Report 01

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Abstract

This is our teams abstrace we may talk about Brief overview of objectives, methodology, and key findings at this section.

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1.1 Introduction

1.2 Objective

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2. Axial Compression Test (*Essai de Compression Axiale*)

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3. Local Compression Test (*Essai de Compression Locale*)

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4. Shear Test (*Essai de Cisaillement*)

4.1 Introduction

Shear stress in timber occurs when forces act parallel to the grain direction, causing one layer of fibers to slide over another along a plane parallel to the grain. This type of failure is particularly critical in notched beams, connections, and lap joints.

In timber structures, shear failure (cisaillement) is one of the most sudden and brittle failure modes, as wood has significantly lower shear strength parallel to grain compared to its tensile or compressive strength. Understanding shear behaviour is therefore essential for safe structural design.

The design shear stress must satisfy:

$$\frac{\tau_d}{f_{v,d}} \leq 1$$

If the beam has a notch (entaille) at the support:

$$\frac{\tau_d}{k_v \cdot f_{v,d}} \leq 1$$

4.2 Objective

The objective of this shear test are as follows:

1. To determine the shear stress ($\tau_{v,d}$) of each timber specimen parallet to the grain.
2. To compute the mean shear stress (τ_m), standard deviation ($\bar{S}$), and coefficient of variation (C_v).
3. To determine the characteristic shear resistance $f_{v,k}$ using Method 1 (Wood Handbook): $\sigma_{b12,m} = \tau_m - S_m$, then $f_{v,k} = \sigma_{b12,m} \times p_k$ (for shear $p_k = 14\%$).
4. To verify the result using formula: $f_{v,k} = 0.2 f_{m,k}^{0.8}$.
5. To observe the shear failure mode and compare experimental result with standard design values.

4.3 Calculations and Results

4.3.1 Sample Dimensions

Three timber lap joint sample were tested on 26 February 2026 using the Machine at ITC lab. The samples are the stepped lap joint geometry. The shear plane area A is defined as the width b multiplied by the shear height h'.

No.	$a(mm)$	$b(mm)$	$h(mm)$	$h' + h(mm)$
1	20.03	29.95	16.18	28.03
2	20.18	30.66	16.65	28.14
3	20.09	29.92	16.29	28.00

Table 4.1: Dimension of shear testing sample

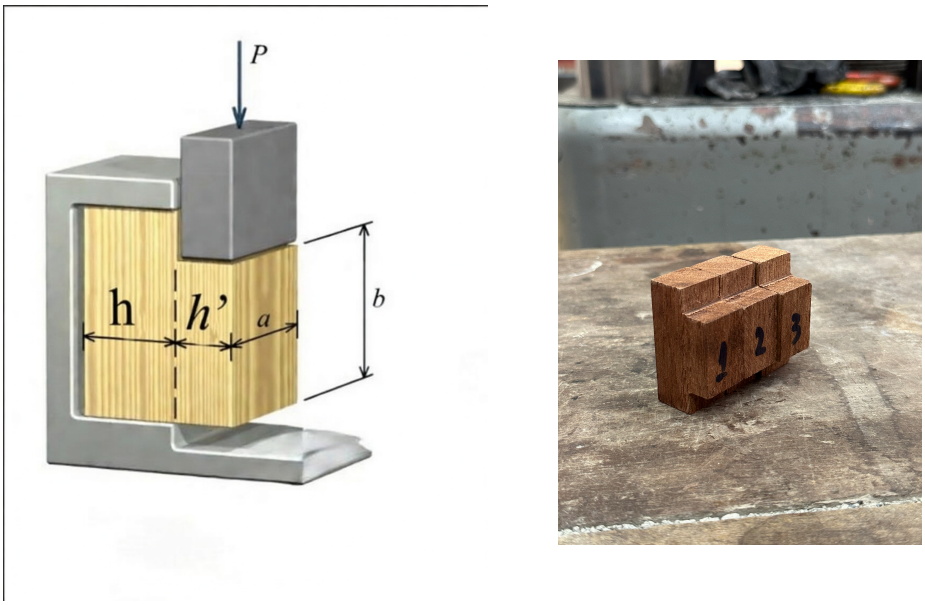

Figure 4.1: Shear testing's sample

4.3.2 Machine Test Results

The following results were recoreded directly from the testing machine.

Test No.	ChNo	a (mm)	b (mm)	S (mm²)	Fm (kN)	Rm (MPa)
5215	N° 1	20.03	29.95	599.9	9.933	16.56
5216	N° 2	20.18	30.66	618.7	8.513	13.76
5217	N° 3	20.09	29.92	601.1	7.707	12.82
Mean	—	20.10	30.18	606.6	8.718	14.38

Table 4.2: Machine test results

4.3.3 Shear Stress Calculation

For each test piece, we calculate the shear stress at failure (rupture) using the formula:

$$\tau_{v,d} = \frac{F_m}{A} = \frac{F_m}{a \cdot b}$$

where F_m is the maximum applied load (N).

Sample 1 (Test 5215):

$$A_1 = b \times a = 29.95 \times 20.03 = 599.9mm^2$$
$$\tau_1 = \frac{9.933[kN]}{599.9[mm^2]} = 16.558MPa$$

Sample 2 (Test 5216):

$$A_1 = b \times a = 30.66 \times 20.18 = 618.719mm^2$$
$$\tau_2 = \frac{8.513[kN]}{618.719[mm^2]} = 13.759MPa$$

Sample 3 (Test 5217):

$$A_1 = b \times a = 29.92 \times 20.09 = 601.093mm^2$$
$$\tau_2 = \frac{7.707[kN]}{601.093[mm^2]} = 12.822MPa$$

4.3.4 Statistical Analysis

Mean shear stress:

$$\tau_m = \frac{\sum_{i=1}^{n} \tau_i}{n}, \quad (n = 3)$$

then,

$$\tau_m = \frac{16.558 + 13.759 + 12.822}{3} = 14.38MPa$$

Standard deviation:

$$\bar{S} = \sqrt{\sum_{i=1}^{n} \frac{(\tau_i - \tau_m)^2)}{n-1}}, \quad (n = 3)$$

then,

$$\bar{S} = \sqrt{\frac{(16.558 - 14.38)^2 + (13.759 - 14.38)^2 + (12.822 - 14.38)^2}{3-1}} = 1.944MPa$$

Coefficient of Variation (C_v):

$$C_v = \frac{\bar{S}}{\tau_m} \times 100\% = \frac{1.944}{14.38} = 13.518\%$$

4.3.5 Characteristic Shear resistance

4.4 Conclusion

5. Bending Test (*Essai de Flexion*)

5.1 Introduction

5.2 Objective

5.3 Calculations and Results

5.4 Conclusion

General Conclusion

This part may be add if we think it is needed.

References

[1] Let say it is References 1

[2] Let say it is References 2 ...

Appendices

This is appendices